SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Security Attacks, Threats, Risks and Their Impact on Security Goals

Technical Presentation

PRESENTED BY

- Dubba Naveen — 192372177

Security Goals: The CIA Triad

Confidentiality	Integrity	Availability
Only authorized users can access information.	Data stays accurate and is not altered without permission.	Systems and data are accessible when needed.

Related goals: Authentication, Non-repudiation, Accountability

Threat vs. Vulnerability vs. Risk

Threat	A potential event or actor that can cause harm. *e.g. hacker, malware, insider*
Vulnerability	A weakness that a threat can exploit. *e.g. unpatched software, weak password*
Risk	Likelihood of a threat exploiting a vulnerability, times the impact. *Risk = Threat x Vulnerability x Impact*

Types of Security Attacks

Passive Attacks

Goal: observe or steal information without altering it

- Eavesdropping / sniffing
- Traffic analysis
- Hard to detect, so focus on prevention

Active Attacks

Goal: modify, disrupt or fabricate data and services

- Masquerading / spoofing
- Replay and man-in-the-middle
- Modification of messages
- Denial of Service (DoS/DDoS)
- Easy to detect, hard to prevent

Impact of Attacks on Security Goals

Attack	Example	Goal Violated
Snooping / Sniffing	Reading unencrypted traffic	**Confidentiality**
Modification	Altering a bank transfer amount	**Integrity**
Denial of Service	Flooding a server with requests	**Availability**
Masquerading	Pretending to be a valid user	**Authentication**
Repudiation	Denying a sent transaction	**Non-repudiation**

Managing Risk: Countermeasures

Prevent

Encryption, firewalls, strong authentication, patching

Detect

IDS/IPS, logging, monitoring, audits

Respond

Incident response plan, backups, recovery

Assess

Regular risk assessment and security awareness training

Key Takeaways

1. Threats exploit vulnerabilities, and risk measures the resulting impact.
2. Every attack targets one or more goals of the CIA triad.
3. Layered defense: prevent, detect, respond.

Thank you. Questions?